The purpose of this document is to propose a collaborative LLM-aided attack on the central problem of constructive quantum field theory: the construction of Yang–Mills theory in four spacetime dimensions (YM_4). This is arguably the least difficult component of the Yang–Mills millennium problem, but also the most important (in my opinion). The mass gap is likely the hardest, followed by a proof of the Wightman/Osterwalder–Schrader axioms. The difference is that the mathematical literature already contains a strategy for attacking existence, developed by Bałaban almost 40 years ago [Bał85a; Bał85b; Bał87; Bał88b; Bał88a; Bał89]. The upshot of the proposal below is that interested mathematical physicists should combine our efforts to see how far LLMs can push Bałaban's program. If we are lucky and the LLMs are capable of producing a formally-verified proof of Yang–Mills existence, then we should combine our efforts to produce something understandable by humans. Whatever fruits the LLMs produce, I think it is best if we enjoy them together. I do not know what the chances of success are — I have no special knowledge here (except for the results of experiments described below) — but even if the probability of success is low, the reward is large, so this seems worth pursuing.

Bałaban's work (cf. also [MRS93]) is widely acknowledged to contain the seeds of a solution to the existence problem for Yang–Mills in finite volume. The issue is that carrying that program to completion would require superhuman levels of work. Even Bałaban himself is merely human. At some point, he decided that his approach was too difficult. Rather than carrying it out, he preferred to wait for some conceptual advance to flatten the difficulty landscape. No conceptual advance came, and four decades later the situation is much the same as where Bałaban left it in 1989. The old-school constructive quantum field theorists redirected their attention to more tractable problems in statistical mechanics and low-dimensional quantum field theory, and the mathematical physics community moved on. Not because of some fundamental obstacle, but because the amount of work required was thought to be more than humans are capable of.

It took four decades, but now the difficulty landscape *has* changed, not because of some new conceptual advance, but because of the advent of artificial intelligence in the form of LLMs. Following Bałaban's program requires a superhuman amount of work, *but AI is capable of that.* In fact, this ought to be an ideal use case for AI – not as a replacement for human ingenuity and understanding, but as an enhancement. We know how far Bałaban was able to go, but not Bałaban+AI. It could well be that the answer is all the way. Why not see?

As an initial experiment, inspired by the seemingly exponential progress of frontier models at research level mathematics and a talk given by Wojciech Dybalski [DST26] at a conference in Bedlewo to which I was invited, I asked Codex to try to reproduce Dimock's work [Dim15; Dim18; Dim20; Dim22] on QED3. In those papers, Dimock attempted to digest and apply the core elements of Bałaban's method to the more tractable problem of quantum electrodynamics in three spacetime dimensions (QED3). After several days, the model (5.6 Sol) reported success. I then asked it to try and establish the convergence of correlation functions, the existence of the infinite-volume limit, and the basic axioms. After about two more weeks of compute, with a small number of agents working in parallel, the model reported success. I asked the model to write a human-readable summary. I do not know whether the manuscript is correct. I did my best to have another model referee it. (I should probably have done a more thorough job.) The experiment was a success as far as I was concerned, not because I was confident that the manuscript was completely correct (I am not), but because it seemed to show that the models were by then capable of a serious attack.

Buoyed by something that wasn't a complete failure, I emailed Levent Alpöge, whom I recognized as having an excellent track record of getting the models to produce correct results. In that email, I suggested attacking Yang–Mills along Bałaban's line. This repository contains the results of that attempt (still in progress). These are raw LLM outputs, not manuscripts or anything meant to be read by a human, just what the machine spat out at this stage, with some small amount of direction. Levent did and is continuing to do a fantastic job of coaxing Claude, but the result leaves a lot to be desired. There are certainly gaps and mistakes. I cannot tell how reliable the reported progress is.

We all know that LLMs have a tendency to be overly optimistic. But I strongly believe it is worth seeing how much further this can go, and the LLM output has not stabilized yet. I would assume that many other people have been trying this in parallel, so this repository records our particular attempt, as of Oct. 6th, 2026.

My motivation here is selfish: I want to understand QFT. The whole reason I became a mathematician was because, as a budding physicist, I was scandalized by physicists' treatment of QFT. If we can now extract understanding of QFT from AI, there is nothing I would rather be doing mathematically. Ultimately, I had hoped that Bałaban, Dimock, or some other constructive quantum field theorist would suggest a working group with the goal of the production and digestion of an AI-written manuscript on Yang–Mills. This didn't happen. Regardless, this seems like something worth doing. Thus, I propose preemptively forming a working group for the production, formalization, digestion, and rewriting of an AI-written manuscript on Yang–Mills.

Even if OpenAI or some other company produces a solution, my proposal re. digestion remains unchanged.

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